



Model T750 Portable Gas Calibrator

The Model T750 is a portable, microprocessor based calibrator for precision gas analyzers. Using highly accurate mass flow controllers combined with compressed sources of standard gases, calibration standards are provided for multipoint span and zero calibrations. The T750 is housed in a high-density, molded travel case which includes an integral handle and wheels for durability and ease of use in the field.

The Model T750 can be equipped with an optional UV ozone generator for accurate, dependable ozone calibrations, and production of NO₂ when blended with NO gas in the included, internal gas phase titration (GPT) chamber. An optional photometer allows even more precise control of the ozone generator giving accurate and dependable ozone calibrations and enhanced GPT. A number of calibration sequences may be programmed into the non-volatile memory of the Model T750, covering time periods up to a year.

All T Series instruments offer an advanced color display, capacitive touch screen, intuitive user interface, and flexible I/O. All instrument set up, control and diagnostic information is available through the front panel, or via RS232, Ethernet, or the optional USB com port, either locally or by remote connection using the included APIcom™ software.

Free Customer Support by telephone and email for the life of the instrument

- » Generates precise calibration gases for SO₂, H₂S, NO, NO₂, CO, O₃ and others
- » Large, vivid, and durable color graphics display with touch screen interface
- » Ethernet, RS-232, and (optional) USB com ports
- » Front panel USB ports for peripheral devices and firmware upgrades
- » Nested sequences (up to 5 levels)
- » Software linearization of mass flow controllers (MFC)
- » 3rd MFC for wide dynamic range (optional)
- » Gas phase titration chamber (GPT) (optional)
- » Ozone generator and photometer allow use as a primary or transfer standard (optional)
- » Inlets for external ozone reference sources
- » Two-year warranty

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Specifications

Dilution System

Flow Measurement Accuracy:	± 1% of full scale
Repeatability of Flow Control:	±0.2% full scale
Linearity of Flow Measurements:	±0.5% full scale
Flow Range of Dilution Air:	0 to 10 SLPM
Optional Ranges:	0 to 20 SLPM, 0 to 5 SLPM
Flow Range of Cylinder Gases:	0 to 100 cc/min
Optional Ranges:	0 to 50 cc/min, 0 to 200 cc/min
Zero Air Requirements:	10 SLPM @ 30 psi
Optional:	20 SLPM @ 30 psi
Input Pressure:	20-40 psig
Calibration Gas Input Port:	1
Diluent Gas Input Port:	1

Ozone Generator Module Option

Maximum Output:	5.5 ppm LPM
Minimum Output:	100 ppb LPM
Response Time:	180 seconds to 98%

UV Photometer Option

Range:	0-100 ppb to 0-10 ppm (selectable)
Precision:	1.0 ppb
Linearity:	1.0% of reading
Rise/Fall Time:	<20 seconds (photometer response)
Response Time:	180 seconds to 95% (system response)
Zero Drift:	<1.0 ppb/24 hours

Electrical Specifications

Power Requirements:	85V-264V, 47Hz-63Hz
Analog Output Ranges:	10V, 5V, 1V, 0.1V (selectable)

Communications Specifications

Included I/O:	1 x Ethernet: 10/100Base-T 2 x RS232 (300-115,200 baud) 2 x USB device ports 12 x digital control outputs 12 x digital control inputs 8 x digital status outputs
Optional I/O:	1 x USB com port 1 x RS485 Multidrop RS232

Physical Specifications

Operating Temperature Range:	5 - 40°C
Dimensions (HxWxD):	9" x 17" x 21" (229 x 432 x 533 mm)
Weight:	37 lbs (16.78 kg) 46.6 lbs (21.14 kg) with photometer, GPT, and O ₃ generator

How to Order

Model T750 includes:

- ☐ Two year warranty
- ☐ 1 calibration gas inlet port
- ☐ 1 diluent gas inlet port
- ☐ 12 digital control outputs
- ☐ 12 digital control inputs
- ☐ 8 digital status outputs
- ☐ RS-232 ports
- ☐ Ethernet port
- ☐ USB ports for peripheral devices
- ☐ APIcom™ remote control software
- ☐ Select AC input voltage
 - ☐ 100V - 120V ☐ 50Hz
 - ☐ 220V - 240V ☐ 60Hz
- ☐ Select DC output voltage
 - ☐ 10V ☐ 5V
 - ☐ 1V ☐ 0.1V

I/O Options:

- ☐ USB com port
- ☐ Multi-drop RS232
- ☐ RS485

Other Options:

- ☐ O₃ generator and GPT mixing chamber
- ☐ UV photometer module
- ☐ Photometer certifications

The values expressed above are in accordance with EPA definitions, where applicable.
All error specifications are based on constant conditions.
Specifications subject to change without notice.
Printed documents are uncontrolled. SAL000085E (DCN 7092) T750/5.14.15



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For more information about the Teledyne API family of monitoring instrumentation products, call us or visit our website at

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